M378K: April 6th, 2026.

Confidence intervals for the μ w/ variance unknown.

Focus on the normal model $N(\mu, \sigma)$ w/ both parameters unknown, but, w/ target parameter μ.

Theorem.

- $\bar{Y} \sim \text{Normal}\left(\text{mean} = \mu,\ \text{sd} = \frac{\sigma}{\sqrt{n}}\right)$
- $Q^2 = \frac{1}{\sigma^2} \sum_{i=1}^{n} (Y_i - \bar{Y})^2 \sim \chi^2(df = n-1)$
- $\bar{Y}$ and Q^2 are independent.

$S^2(n-1) = \sigma^2 \cdot Q^2$

Goal: Confidence interval for μ.

Idea:

$$\boxed{U = \frac{\bar{Y} - \mu}{\frac{S}{\sqrt{n}}}} \quad \text{w/} \quad \boxed{S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2}$$

pivotal quantity ✗

$$\frac{\frac{\bar{Y} - \mu}{\frac{\sigma}{\sqrt{n}}}}{\frac{S}{\sigma}} = \frac{Z}{\sqrt{\frac{S^2}{\sigma^2}}} = \frac{Z}{\sqrt{\frac{Q^2}{n-1}}}$$

$\sim N(0,1)$
$\sim Z$

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t-Distribution.

Def'n. A student t-distribution w/ k degrees of freedom is the dist'n of the r.v.

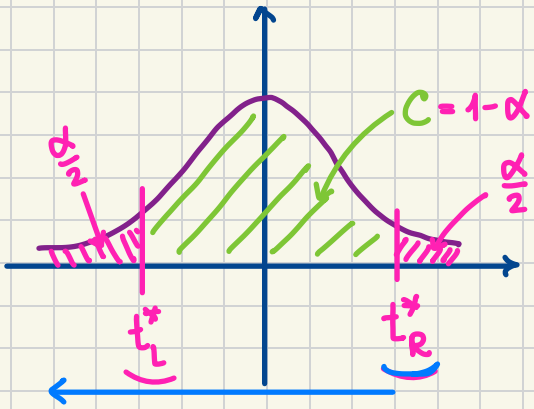
$$T = \frac{Z}{\sqrt{\frac{Q^2}{k}}}$$

w/
- $Z \sim N(0,1)$
- $Q^2 \sim \chi^2(df = k)$
- Z and Q^2 are independent

We write: $\boxed{T \sim t(df = k)}$

Back to Confidence Intervals.

To construct a confidence interval w/ the confidence level $C = 1-\alpha$, we do



$$t^* = t^*_R = -t^*_L$$

$1-\alpha+\frac{\alpha}{2} = 1-\frac{\alpha}{2}$

$C + \frac{1-C}{2} = \frac{(1+C)}{2}$

$$t^* = qt(1-\alpha/2, df=n-1)$$

$$t^* = qt((1+C)/2, df=n-1)$$

$$\mathbb{P}\left[-t^* \leq \frac{\bar{Y}-\mu}{\frac{S}{\sqrt{n}}} \leq t^*\right] = C = 1-\alpha$$

$$\mathbb{P}\left[-t^* \cdot \frac{S}{\sqrt{n}} \leq \bar{Y}-\mu \leq t^* \cdot \frac{S}{\sqrt{n}}\right] = 1-\alpha$$

$$\mathbb{P}\left[\underbrace{\boxed{\bar{Y} - t^* \cdot \frac{S}{\sqrt{n}}}}_{\hat{\theta}_L} \leq \mu \leq \underbrace{\boxed{\bar{Y} + t^* \cdot \frac{S}{\sqrt{n}}}}_{\hat{\theta}_R}\right] = 1-\alpha$$

$$\boxed{\mu = \underbrace{\bar{Y}}_{\text{pt. estimator}} \pm \underbrace{t^*}_{\text{critical value}} \underbrace{\frac{S}{\sqrt{n}}}_{\text{std error}}}$$

Problem 16.9. *(20 points)*
A random sample of size 10 is drawn from a normal distribution with **both** *mean and standard deviation* **unknown.** *The generated sample has the sample mean $\bar{y}_{10} = 14$ and the (unbiased) estimate of the variance $s^2 = 25$.*

(i) (10 points) Construct a (symmetric) 90%-confidence interval for μ.

(ii) (10 points) Construct a (symmetric) 90%-confidence interval for σ^2.
Hint: *Remember that you know the distribution of $(n-1)S^2/\sigma^2$.*

$$\mu = \bar{Y} \pm t^* \cdot S/\sqrt{n}$$

$$= 14 \pm t^* \cdot \frac{5}{\sqrt{10}}$$

$C = .9 \Rightarrow \alpha = .1$

$t^* = qt(1 - \frac{.1}{2}, df = 9)$

$= .95$

$$\mathbb{P}\left[\frac{(n-1)S^2}{b} \le \sigma^2 \le \frac{(n-1)S^2}{a}\right]$$

$$\left(\frac{9 \cdot 25}{b}, \frac{9 \cdot 25}{a}\right)$$

$a = qchisq(\frac{.1}{2}, df = 9)$

$b = qchisq(1 - \frac{.1}{2}, df = 9)$